



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - ATSB-style final report

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INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

At 18:47 UTC on 11/08/2026 (02:47 local time), haul truck HT-407, a 220 t class rigid haul truck descending empty on the Stage 4 open pit main ramp at Bindarra Gold Mine, struck the right side of light vehicle LV-118 as it emerged from the RL 312 dewatering spur. The spur had been cut under a works order and first used by light vehicles on 02/08/2026. Coming over a crest, the truck operator sighted the light vehicle at about 40 m. Within 1.4 seconds he braked and steered right towards the inner windrow, reducing speed from 37 km/h to 24 km/h at impact. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow.

The driver, a grade control technician employed by a survey subcontractor, was on his third night shift at the mine. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated through the rear of the vehicle 27 minutes after the collision and was flown to the metropolitan trauma centre. His seatbelt was fastened. The roll-over protection structure deformed but was not breached. The light vehicle was destroyed. The haul truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit was stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours.

What the investigation found

The spur joined the ramp 34 m beyond a crest. This gave the truck operator 41 m of sight line, against a plan minimum of 120 m and a stopping distance of 96 m at the posted speed. Neither operator was aware of the other.

The works order was classified as routine earthworks, and no management of change was raised. This very likely contributed, because it removed the only route by which the new intersection would have reached the traffic management plan owner, the in-cab map and the collision awareness system configuration. Eleven of the fourteen new Stage 4 roads cut since the last plan review had been built the same way.

Positive radio calls were required only at nine listed intersections, and the list could change only at the annual review. This very likely contributed. Neither operator made a call. The collision awareness system detected the light vehicle but suppressed the alert because a loading area geofence, extended in June 2026, covered the intersection. The alert would have come after braking began. Months without any alert on that ramp likely contributed by leading the operator to read silence as a clear road.

The driver entered the ramp without stopping. He had not completed the site's pit driving assessment, which the gatehouse had no means of verifying. The state guideline leaves the review trigger and interaction zones to each site. This likely contributed, although it was held with less confidence than the site's own controls.

The single night shift paramedic and the fatigue exposure of a tenth consecutive night shift did not contribute, but each increased risk.

Whether dust affected visibility could not be determined. A water cart pass was overdue, and particulate levels were elevated but below the site trigger. However, the drivers' accounts differed, the sighting distance matched the clear-air geometry, and nothing at the site measures visibility. A wet surface and degraded brakes were ruled out.

What has been done as a result

At 05:10 local that night, the Mining Manager directed that a positive call be made before entering any Stage 4 intersection, listed or not. The spur was barricaded and closed. No safety action has been recorded in response to the safety issues identified.

Safety message

A new road cut into an operating pit changes the traffic system, however routine the earthworks. Every control that depends on knowing where intersections are, including plans, in-cab maps, radio call lists and collision awareness geofences, should be updated before the road opens to traffic. Where sight distance is short, driver attention alone cannot provide a safe margin.

The occurrence

Recorded data from the haul truck and the pit radio channel are timed in UTC; 18:47:12 UTC corresponds to 02:47 local time, so local times are 8 hours ahead. [NOT ESTABLISHED: the local calendar date of the collision]

The RL 312 dewatering spur

The RL 312 dewatering spur is a 240 m access road in the Stage 4 open pit at Bindarra Gold Mine, cut to reach dewatering bore DW-14 under works order WO-26-4471. The works order classified the job as routine earthworks within the approved Stage 4 design, stated that management of change was not required, recorded completion on 31/07/2026 and entered the handover field as "N/A".

On 02/08/2026 the bore was energised and light vehicles first used the spur. The works log recorded that the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed that day and not replaced. At the time of the collision, the intersection had no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed. It did not appear on the traffic management plan road layout drawing or on the fleet management system (FMS) in-cab map, last published on 24/07/2026.

The spur joins the main ramp from the east, on the inside of a left-hand bend, 34 m beyond the high point of a crest. The survey found that:

- a descending haul truck operator, at an eye height of 5.80 m, first had line of sight to the spur mouth at 41 m
- a light vehicle driver at the spur mouth first had line of sight to a descending truck at 38 m
- the 1.6 m inner windrow obscured the truck's lower body before that point.

At 22:10 local on 04/08/2026, seven days before the collision, a dewatering technician submitted hazard report HAZ-2026-0917. He wrote that on driving out of the spur he found a haul truck descending "maybe sixty metres up"; neither driver had seen the other until he was on the road, and he reversed into the spur to let it pass. He wrote that "you cannot see up the ramp from that road", citing the rise and the windrow, and noted the absence of a warning sign for trucks and of a give-way control.

The Dewatering Superintendent closed the report on 06/08/2026, recording that the crew had been reminded at pre-start to take care entering the main ramp and to give way to haul trucks, and reduced the risk rating from Medium to Low. The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner.

The collision

At 18:22:41 UTC (02:22 local) dispatch called the Stage 4 crews back after crib. HT-407, a 220 t class rigid haul truck, reported "407 dumped, coming back down" at 18:38:55 and descended the main ramp empty. The FMS recorded 41 km/h at RL 368 at 18:45:58, 39 km/h at RL 348 with the retarder on the second position, and 37 km/h at 18:47:06 at RL 314, 41 m above the spur intersection.

The driver of light vehicle LV-118, a grade control technician employed by a survey subcontractor on his third night shift at Bindarra, had completed levels at the DW-14 bore collar and drove back towards the ramp. His account was that:

- it was his second visit to the bore, and he had driven the spur six or seven times without meeting a haul truck

- he slowed at the spur mouth, looked left up the ramp, saw nothing and drove out at about 25 km/h
- he first said he had stopped at the mouth, then asked that the notes record he did not believe he had come to a complete stop
- his beacon was on and his seatbelt fastened.

He recalled leaving the collar at about ten to three, later than the recorded collision time; the time he left was not established from recorded data.

Neither driver made a radio call. The traffic management plan requires a positive call at nine listed intersections, which did not include the RL 312 spur. The night shift supervisor did not know a light vehicle was in the pit, and stated there was no permit, call-in or board by which he could have known.

The truck operator knew the spur had been cut and had noticed the closed sign was gone, but believed the road was still closed. His in-cab display showed no intersection at RL 312. He stated that in fourteen months the collision awareness system had never alerted him anywhere in the pit, and he had taken that to mean the road was clear.

The ramp surface was dry and well maintained. The truck operator said some dust was hanging but could not say whether it made a difference; the light vehicle driver said the air was clear. The truck operator reported seeing the light vehicle on the ramp at about 40 m as he came over the crest, consistent with the surveyed 41 m sight line.

At 18:47:07.4, 1.4 seconds after the last steady data point, service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied; the operator stated he steered right into the windrow because the left side was a drop. The braked tyre scuff began 38.5 m up-ramp of the spur mouth and deviated 1.4 m right over its first 20 m.

At 18:47:09.2 the collision awareness system logged LV-118's tag at 21.4 m, closing at 13 km/h. The event fell inside the S4-LOAD suppression geofence, so no alert was presented. The system operated at Level 7 (operator advisory); Level 9 (automated intervention) was available on this fleet but not enabled at Bindarra.

At 18:47:12.0 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m from the inner windrow toe. Mean deceleration from brake application was 1.85 m/s², against 1.9 m/s² in the brake test on 03/08/2026. Without braking, collision speed would have been 37 km/h.

The truck pushed LV-118 18.4 m down the ramp; it went over at 11.2 m and came to rest on its left side against the inner windrow, facing down-ramp. HT-407 stopped at 18:47:14.3, 19 m below the intersection, with damage confined to the front left bumper, left mudguard and lower two access ladder steps.

LV-118 was destroyed, with damage concentrated on the right side from the B pillar forward. Its roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened, with loading marks on the webbing. The beacon whip had been sheared at its base; its lamp worked when tested after recovery.

Emergency response and immediate actions

At 18:47:41 the truck operator called: "Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve." At 18:48:01 the dispatcher called: "All Stage 4 units stop. All units stop." The operator was told to stay in the cab until traffic had stopped and was released at 18:48:44.

The supervisor headed to the scene from the ROM pad and, on the operator's report that the driver was conscious but trapped with the door against the ground, instructed that he not be moved.

The emergency response team was paged at 02:48 local; the first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The night shift was rostered with one mines rescue paramedic, and a second responder must be present before any extrication. The second responder was called from the accommodation village, 11 km from the gatehouse, and arrived at 03:07 with the hydraulic rescue tool.

The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, reached the surface medical facility at 03:41 and was flown to the metropolitan trauma centre at 07:40. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. The post-incident review found extrication was delayed by approximately 9 minutes awaiting the second responder and tool, and concluded this did not affect the clinical outcome because the injuries were stable and the patient was conscious throughout.

After the collision:

- the Stage 4 pit was stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours
- the spur was barricaded and closed to all traffic
- both vehicles were left in position and the scene was surveyed
- the truck operator and shift supervisor were stood down and tested for drugs and alcohol, with negative results
- FMS and collision awareness system data were downloaded at 09:30 local
- at 05:10 local the Mining Manager directed that a positive call be made on the pit channel before entering any Stage 4 intersection, whether listed in the plan or not
- HT-407 returned to service on 14/08/2026
- the occurrence was notified to the State Mines Inspectorate.

Context

Personnel information

Haul truck operator

The truck operator had 11 years' open pit mining experience, including seven years on 220 t class rigid haul trucks across three sites. He had been employed by Warrigal Mining Services for 4 years 2 months, 14 months of it at Bindarra, and held current verifications of competency on the 220 t class unit (issued 06/2025), dozers and water carts. He completed his annual refresher in 02/2026.

He completed the site general induction in 06/2025 and had no traffic management refresher since. The induction's 23 slide traffic management module named the nine intersections listed in the traffic management plan but did not address unlisted intersections. No training record on site, for any person, covered the collision awareness system's suppression geofences.

He worked a 14 night on, 7 off roster from 18:00 to 06:30 local, a 12 hour 30 minute shift with a 30 minute crib break. The collision occurred on night 10 of 14, 10 hours 47 minutes into the shift. Site access records confirmed the previous nine shifts were worked in sequence, none beyond 12 hours 40 minutes, and the contractor's fatigue limits of 14 consecutive night shifts and a minimum 10 hour break were met. His voluntary sleep record showed 6 hours 30 minutes of sleep the day before the occurrence and between 5 and 7 hours on each of the preceding nine days.

Light vehicle driver

The light vehicle driver was a grade control technician employed by Merrivale Survey & Geotechnical since August 2022, with 11 years' survey experience, six in mining, rotating between four client sites on a 9 on, 5 off roster from 18:00 to 06:00. Bindarra was new to him; he worked night shifts there on 09/08, 10/08 and 11/08.

His training file recorded the Merrivale general mining induction, a two hour online module completed on 24/07/2026. His Bindarra site access card was issued on 09/08/2026 against Merrivale's recognition agreement, after a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. There was no evidence he had been issued, completed or signed for the Bindarra site-specific supplement or the Bindarra light vehicle pit driving assessment; the gatehouse neither required nor asked for either. He told the investigation he had not heard of the pit driving assessment and had never seen, or known of, the Bindarra traffic management plan.

The dewatering contractor's foreman had driven him to the spur on his first night, told him it was open to light vehicles and, according to the driver, which radio channel to use. The driver said nobody had told him to call before entering the ramp.

Night shift supervisor

The night shift supervisor had 19 years' open pit mining experience, 8 years with Warrigal Mining Services and 2 years 6 months supervising the Stage 4 night shift. He completed the site general induction in 04/2024 and a supervisor refresher in 11/2025, and worked the same night 10 of 14 pattern as his crews.

He was responsible for the haul fleet, excavators, ancillary fleet, dozers and dispatch operator, but not the dewatering crew, the survey contractor or other light vehicles in the pit. He believed Warrigal supervisors had at some point been trained on BIN-TMP-004, as distinct from Warrigal's own procedures, but no training evidence was located.

Vehicle information

HT-407

HT-407 was a 220 t class rigid haul truck commissioned in 03/2021, with 31,480 engine hours. Its last 500 hour service, on 28/07/2026, carried forward no defects. The monthly brake test on 03/08/2026 passed with a mean deceleration of 1.9 m/s² empty on a 9 per cent grade; a test on 14/08/2026, before return to service, gave 1.92 m/s². The retarder, service brakes and park brake met specification before and after the occurrence. The 11/08/2026 pre-start check sheet was complete with no defects, and none had been raised against lighting, mirrors, cameras or cab glass in the preceding twelve months.

LV-118

LV-118 was a 4WD utility in service since 02/2024, with 61,200 km, last serviced on 17/07/2026, 4,100 km before the occurrence. It had a certified roll-over protection structure, a 4.5 m amber beacon whip, reflective marking to the site standard and a UHF radio, and its inspection sheet for the week of 09/08/2026 recorded all items serviceable. LV-118 was not dispatched and carried no FMS. Its collision awareness system transponder was a passive tag, so its movements were known only from the proximity log, the driver's account and the scene evidence.

Collision awareness system

Advisory alerts were raised when a vehicle tag entered a 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. Because the alert rate had made the system unusable, the S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face to stop nuisance alerts between the excavator, trucks and nearby service vehicles.

On 09/06/2026 the contractor's technical services group extended the geofence northward as a system configuration task, placing its boundary 180 m up-ramp of the RL 312 intersection. The extension was not recorded in, or reviewed against, the traffic management plan. The night shift supervisor had learned in about the middle of 2025 that alerts were switched off around the loading face, but did not know of the extension.

Environmental conditions

The collision occurred in darkness. The Stage 4 pit had no fixed lighting; the loading face light plants were 1.1 km away and out of sight behind the crest. Moonset had been at 23:18 local the previous evening.

The Bindarra met station at the ROM pad recorded wind from the south-east at 8 km/h, a temperature of 9 degrees and no precipitation; the last recorded rainfall was 26 days earlier.

The water cart log recorded the main ramp between RL 280 and RL 400 as last watered at 17:10 UTC, 1 hour 37 minutes before the occurrence, making a pass 7 minutes overdue against the standing instruction of one every 90 minutes.

A fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre for 18:40 to 18:50 UTC, against a shift mean of 88 and a site trigger of 250. This measured particulate at the monitor head only. Nothing at Bindarra measured visibility, and no measurement existed of what could be seen from either cab on the night.

Regulatory guidance

Mines Safety Guideline MSG-14, revision 4 (2021), was issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018, which requires only that

"the operator shall manage the risks arising from the interaction of vehicles at the mine". Compliance with the guideline is one way of meeting that requirement. Under the guideline:

- **Clause 3.2** requires review of the traffic management plan at intervals not exceeding twelve months and "following any significant change to the mine's road network, vehicle fleet or operating pattern", without defining significant change.
- **Clause 3.5** makes the plan a site document, not approved by or submitted to the Inspectorate except on request.
- **Clause 5.1** requires nomination of interaction zones requiring positive communication, with no minimum and no basis for nomination.
- **Clause 5.4** requires a light vehicle to be detectable by a heavy vehicle operator by line of sight, elevated marking, radio communication, a proximity detection system, or a combination, without specifying which or how many.

On the day of the occurrence, Bindarra's plan met the guideline on its face.

Site procedures

Traffic management plan

BIN-TMP-004 revision 7, owned by the Mining Manager, was issued on 12/11/2025 with the next review due on 12/11/2026. Its main provisions were:

- **Clause 4.1:** light vehicles to "give way to heavy vehicles at all times and in all circumstances".
- **Clause 4.3.1:** a positive call before entering any intersection listed at Appendix B.
- **Clause 4.3.2:** at unlisted intersections "No call is required".
- **Clause 6.2:** available sight distance at every intersection not less than the stopping distance of the largest vehicle at the posted speed, "and in no case less than 120 m", otherwise a stop, signalised or traffic controller control. The stopping distance of a 220 t class truck empty at the posted 40 km/h on a 9 per cent descending grade, from the manufacturer's data and a 1.5 second response, was 96 m.
- **Clause 6.4:** warning signs, a give-way control and delineation at every intersection shown on the Schedule 1 road layout drawing.

Appendix B listed nine intersections. It and Schedule 1, dated 12/11/2025, were reissued only at the annual review, when an intersection on Schedule 1 assessed as requiring positive communication would be added. No provision allowed an intersection to be added between reviews or required a new road to be referred to the plan's owner.

Clause 9.1 provides for review annually and after any significant change to the road network. Annual reviews were completed on 14/11/2023, 10/11/2024 and 12/11/2025; no out-of-cycle review had been held in the plan's eleven year history. The Stage 4 crib room displayed the 12/11/2025 Appendix B on a laminated board.

The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine intersections on the board and nowhere else. The supervisor said he had never corrected anyone for not calling elsewhere.

There was no permit, radio call-in, board or check at the pit entry for light vehicles; anyone with a site access card could drive into the pit.

Management of change

CRL-PRO-022 clause 5.2 requires a management of change record before any change to the road network. Clause 5.3 excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced". Clause 5.4 requires the change owner to refer the change to the traffic management plan owner, the survey department (for the FMS road network) and the contractor's technical services group (for the collision awareness system configuration). These referrals exist only within the management of change record; the Schedule 4 works order form has no equivalent fields.

Contractor management

CRL-STD-009 clause 7.3 allows a contractor's general induction to be accepted as equivalent where the contractor holds a current recognition agreement, as Merrivale had since 2023. Clause 7.4 states that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment", which "shall not be substituted". Clause 7.5 makes the engaging department responsible for verifying this before a card is issued. In practice, the gatehouse issued cards against the recognition agreement alone, and the access card system did not distinguish a person who may drive in an operating pit from one who may not.

Related occurrences and system data

The survey department's road register recorded 14 new roads cut in Stage 4 since the 12/11/2025 review. Three were cut under a management of change record and all were referred to the plan's owner; the other 11 were cut under a works order and none was referred.

A search of Bindarra hazard reports over the 24 months to 11/08/2026 returned four, including HAZ-2026-0917. The other three, from 03/2025, 09/2025 and 02/2026, each described a light vehicle and a haul truck meeting unexpectedly at a different unlisted intersection. All four were closed within three days by the receiving department with a reminder to the crew; none was referred to the plan owner or led to a change to Appendix B, Schedule 1 or any physical control.

In the 12 months to 11/08/2026, HT-407 raised 412 advisory alerts, all outside the S4-LOAD geofence. Across the Stage 4 haul fleet, 68 proximity events inside the geofence were suppressed, nine involving a light vehicle tag on the main ramp below the geofence boundary. Suppressed events were logged and could be queried, but the technical services group's standing report covered only alert counts and system health, not suppressions.

Safety analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14, revision 4, was issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline is one way of meeting that requirement. Its text is documentary and not in dispute, so it is virtually certain that the discretion described below existed.

Three provisions of the guideline are relevant to this occurrence, and each leaves a central term to the site:

- **Clause 3.2** requires the traffic management plan to be reviewed "at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern". The guideline does not define a significant change and gives no example of one.
- **Clause 5.1** requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry". It does not say who makes the nomination, on what basis, or what the minimum is.
- **Clause 5.4** requires a light vehicle to be detectable by a heavy vehicle operator. It allows this to be done by "line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these". It does not say which means, or how many, must be provided.

Clause 3.5 adds that the plan "is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."

On the day of the occurrence, Bindarra's plan, BIN-TMP-004, met the guideline on its face:

- it had been reviewed within twelve months, on 12/11/2025
- it nominated nine interaction zones at Appendix B
- it provided for detectability by line of sight and by elevated marking.

The mine was not in breach of the guideline or the regulation. This analysis does not suggest any failure by the Inspectorate. Its concern is the shape of the discretion the guideline leaves, and how that discretion was taken up at the site.

At Bindarra, all three open terms were filled in by one document, BIN-TMP-004, and that document was reviewed once a year with no outside scrutiny. The plan set its own triggers in the same general terms as the guideline:

- Clause 9.1 repeated the requirement for review after "any significant change to the mine's road network" without defining it.
- The note to Appendix B provided that an intersection is added only where it appears on the Schedule 1 road layout drawing, and only at the annual review.
- No provision allowed an intersection to be added between annual reviews.
- No provision required a new road to be referred to the plan's owner when it was cut.

The document register showed that no out-of-cycle review had been held in the plan's eleven-year history. Since the 12/11/2025 review, 14 new roads had been cut in the Stage 4 pit:

- 3 were cut under a management of change record, and all 3 were referred to the plan's owner
- 11 were cut under a works order, and none of these was referred.

The RL 312 dewatering spur was one of the 11. Over the 24 months to 11/08/2026, four hazard reports described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at four different locations. Each was closed within three days with a reminder to the crew. None reached the plan's owner, and none led to a change to Appendix B, to Schedule 1 or to any physical control.

The argument for influence is that a defined trigger for review and a stated minimum for nominating interaction zones would have given those 11 unreferred roads somewhere to land. Every internal control that failed in this occurrence failed for want of exactly that. Under the guideline, the question of when a new road becomes a significant change, and which intersections need a positive call, was answered entirely inside the mine. It was answered in a document that could only change once a year.

There is a substantial argument against. The guideline did not prevent Bindarra from acting:

- The mine could have held an out-of-cycle review at any time under its own clause 9.1.
- Nothing in MSG-14 prevented it from listing every intersection it had at Appendix B.

On this view, the mine did not fail to act because the guideline was vague. It failed to act because nothing inside the mine ever put the question in front of the plan's owner. The external guidance would then be background, and the organisational arrangements would account for the whole of the failure.

The investigation weighed these two arguments and concluded, on the balance, that the width of the discretion in MSG-14 likely contributed to the occurrence. This conclusion was reached with less confidence than the other contributing factors in this analysis, and it should be read as such rather than as settled. The contrary argument was not dismissed. Rather, the investigation considered that guidance defining a significant change, or setting a basis for nominating interaction zones, would more likely than not have created a point at which a new road such as the RL 312 spur came before the plan's owner. The internal arrangements at Bindarra did not provide such a point.

The discretion left by the guideline led to two organisational matters, which are analysed in their own subsections:

- The RL 312 spur was cut and opened to traffic under a works order with no management of change raised. That subsection is under Organisational Influences.
- Positive radio calls were required only at listed intersections, and the list was rewritten once a year. That subsection is under Risk Controls.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

The Bindarra Management of Change procedure, CRL-PRO-022, requires at clause 5.2 that a management of change record be raised before "a change to the mine's road network". Clause 5.3 exempts "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced."

Works order WO-26-4471 cut and formed 240 m of access road to dewatering bore DW-14 at RL 312. It classified the work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3." It did not include:

- signage
- a survey pickup of the new intersection
- a referral to the traffic management plan
- a request to publish a revised fleet management system road network
- a review of the collision awareness system geofences.

Its handover field, the only place any of these could have been recorded, reads "N/A". The procedure, the works order and the survey department's road register agree with each other. It is therefore virtually certain that the spur was built and opened without a management of change record.

The significance of this lies in what the exemption removed. Under clause 5.4, a management of change record requires the change owner to complete an impact assessment that includes referrals to three parties:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system configuration.

These referrals exist only inside the management of change record. The works order form at Schedule 4 has no equivalent fields. None of the three referrals was made, and the consequences are visible in each area they would have reached:

- **The in-cab map:** the fleet management system download showed that the in-cab road network was last published on 24/07/2026, a week before the spur was cut. On the night of the occurrence it still showed the main ramp with no intersection at RL 312.
- **The traffic management plan:** the spur did not appear on the plan's road layout drawing. As a result, the plan's signage, give-way and delineation requirements at clause 6.4 did not apply to it. When the construction sign was removed on 02/08/2026, nothing replaced it.
- **The geofence configuration:** the collision awareness system configuration was never reviewed against the new intersection.

The road register shows this was not an isolated case. Fourteen new roads were cut in Stage 4 after the 12/11/2025 plan review:

- the three raised under a management of change record were each referred to the plan's owner
- none of the eleven raised under a works order was.

The absence of the record was the means by which the intersection stayed outside the plan, the in-cab map and the geofence review. It in turn gave rise to two other conditions:

- the spur was not listed among the intersections requiring a positive radio call
- the spur met the ramp with 41 m of sight line against a plan minimum of 120 m, with no warning or give-way control on either approach.

Both are analysed under Risk Controls and Local Conditions.

One consideration weighs against this conclusion. A referral is not the same as a control. Had a record been raised, the plan's owner might have considered the spur and still left it unlisted. That possibility cannot be excluded. However, the management of change record was the only route by

which the new intersection could have reached any of the three controls, and its absence left all three untouched. On that basis it is very likely that the absence of a management of change record contributed to the occurrence.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

The Contractor Management Standard, CRL-STD-009, permits a contractor's general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Merrivale Survey & Geotechnical had held one since 2023. The standard then draws a clear limit. Clause 7.4 states that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying this before a site access card is issued.

The standard was sound. The gap lay in how it was applied at the point of issue:

- The gatehouse issued cards against the recognition agreement alone.
- It held no record of the supplement or the assessment and was not asked for one.
- Nothing in the access card system distinguished a person who could drive in an operating pit from one who could not.

The verification required by clause 7.5 therefore had no point in the process at which it could take place.

The driver's records are consistent with this:

- His training file records a two hour Merrivale online module completed on 24/07/2026.
- His card was issued on 09/08/2026 after a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment.
- There is no record of the site supplement or the pit driving assessment.
- He had never seen the traffic management plan, BIN-TMP-004, and did not know it existed.
- The occurrence was on his third night shift at Bindarra.

One point could not be fully resolved. Merrivale's own records of the site supplement were not obtained, and the driver was not asked specifically about it. The standard, the training files and the card register are otherwise documentary. It is therefore extremely likely that the driver drove in the operating pit without having completed either item.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice, including:

- give-way
- positive radio calls
- what to do at a blind intersection.

Without it, the driver entered the ramp from the spur without stopping and without establishing the truck's position. That action is analysed under Individual Actions.

Two considerations weigh against a stronger conclusion:

- The rule he would most likely have been taught, that light vehicles give way at all times, was one he was already applying.
- The intersection's 41 m sight line was, on its own, enough to defeat an attentive driver.

The assessment could not have supplied what the geometry denied. Taking both points into account, it is likely that the absence of the site supplement and pit driving assessment contributed to the occurrence.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet operated at Level 7, operator advisory. Level 9, automated intervention, was available on the fleet but was not enabled at Bindarra. Advisory alerts were raised when a vehicle tag entered the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence.

The S4-LOAD geofence was drawn around the Stage 4 loading face on 14/03/2025. Nuisance alerts between the excavator, the trucks under it and the service vehicles there had made the system unusable, and suppressing them was a reasonable measure. On 09/06/2026, as the face advanced, the contractor's technical services group extended the geofence northward as a system configuration task. The new boundary lay 180 m up-ramp of the RL 312 intersection. The extension was not recorded in the traffic management plan and was not reviewed against it. No training record on site, for any person, covered the geofences.

The configuration and event log came from the system itself and were downloaded sealed. It is virtually certain that alerts were suppressed at the intersection on the night. At 18:47:09.2 the system logged LV-118 at 21.4 m, closing at 13 km/h, with the alert suppressed. Outside the geofence, the same event would have produced an advisory tone and a screen indication in the cab.

The influence of this suppression was argued both ways.

The reading against influence. The event came 1.8 seconds after the operator had begun braking and 2.8 seconds before impact. An advisory tone would have told him what he already knew. The system carried no intervention, and with 41 m of sight line the outcome was fixed before either the sighting or the alert. On that reading, the suppression is a serious matter for the fleet but not a cause of this collision.

The reading the investigation accepted. The influence lay not in the final seconds but in the expectation the system had built.

- In the 12 months before the occurrence, HT-407 raised 412 advisory alerts, every one outside the geofence and none inside it.
- Across the Stage 4 fleet in the same period, 68 proximity events inside the geofence were suppressed.
- Nine of those involved a light vehicle tag on the main ramp below the boundary, which is the same arrangement of vehicles and location as the occurrence.
- These events were recorded and visible to anyone who queried the system, but nobody did. The technical services group's standing report covers alert counts and system health and does not report suppressed events.

The truck operator and the shift supervisor each raised the suppression without being asked. The operator's account indicated he had come to treat the absence of alerts on the ramp as meaning the road was clear, and he carried that expectation into the crest.

On that basis, the suppression likely contributed to both vehicles arriving at the intersection with neither operator aware of the other. This conclusion rests on the operator's expectation, not on the

timing of the alert, and the contrary reading was not eliminated. The matter is not that the geofence existed. It is that its boundary moved 180 m and nobody outside the group that moved it was told.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Clause 4.3.1 of BIN-TMP-004 requires that "a positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B". Clause 4.3.2 provides that at other intersections "No call is required."

Appendix B lists nine intersections. Its note provides that an intersection is added "where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication". Both documents are reissued only at the annual review. Nothing in the plan allows an intersection to be added between reviews, and nothing requires a new road to be referred to the plan's owner. As a result:

- The RL 312 spur appeared on neither document.
- The crib room board in Stage 4 was the 12/11/2025 issue.
- The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine listed intersections and nowhere else.

The plan's text and review history are documentary, and it is virtually certain that this condition existed. At this intersection, the mine's own defence against this interaction was inactive by its own terms. The earliest date the spur could have been listed was 12/11/2026, three months after the collision.

The pattern extends beyond this spur:

- Eleven of the 14 new Stage 4 roads cut since the last review were never referred to the plan's owner.
- No out-of-cycle review under clause 9.1 had been held in the plan's eleven year history.
- Four hazard reports, including HAZ-2026-0917, described light vehicles and haul trucks meeting unexpectedly at intersections not listed in Appendix B. Each was closed with a reminder to the crew and none was referred to the plan's owner.

How the spur came to be outside the plan is discussed under Organisational Influences. The guidance behind the plan's trigger is discussed under External Influences. The operators' own actions are discussed under Individual Actions.

The pit channel extract shows no call by either vehicle about the spur, the ramp or a light vehicle in the pit. Had a call been required, the conditions for it to work were in place: the channel was open, both vehicles were on it, and the dispatcher was tracking the truck closely enough to report "407's about four minutes off you" at 18:46:18. The residual doubt is that a call made but not acknowledged still leaves a driver deciding whether to proceed. Even so, the limitation of positive calls to listed intersections very likely contributed to neither operator making a call before entering the intersection.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift was rostered with one mines rescue paramedic at the surface medical facility. The second responder, who must be present before any extrication begins, was called from the accommodation village 11 km from the gatehouse. The roster, response log and post-incident review agree on the times, and it is virtually certain the arrangement existed. All of the following times are local:

- 02:48: dispatch paged the team.
- 02:54: the first response vehicle reached the scene.
- 03:07: the second responder arrived with the hydraulic rescue tool.
- 03:14: the driver was extricated, 27 minutes after the collision.

The post-incident review found extrication was delayed by approximately 9 minutes. It recorded that this delay did not affect the clinical outcome, because the injuries were stable and the driver was conscious throughout. The collision and the injuries were complete before the team was paged, and it is unlikely that the response time had any bearing on this occurrence or its outcome.

The arrangement is nonetheless very likely important to safety. The driver was conscious, restrained and protected by an intact roll-over structure. With a fuel fire, an unconscious casualty, or an airway requiring access in the first ten minutes, the same nine minutes could produce a different outcome. On night shift, the site's extrication capability is one person plus a drive from the village, and nothing in the arrangement scales with the severity of what has happened.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

The survey of 13/08/2026 placed the spur mouth 34 m beyond the high point of a crest in the main ramp. A haul truck operator descending at an eye height of 5.80 m first had line of sight to the spur mouth at 41 m. A light vehicle driver at the spur mouth, at an eye height of 1.45 m, first had line of sight to a descending truck at 38 m. For the last 25 m of the spur's approach, the 1.6 m inner windrow hid the truck's lower body. It is virtually certain that this sight line existed. It was measured to the eye heights of both vehicles, and a dewatering technician had independently described it seven days earlier.

The traffic management plan, clause 6.2, requires the sight distance at every intersection to be at least the stopping distance of the largest vehicle at the posted speed. In no case may it be less than 120 m. Where that cannot be achieved, the clause requires a stop, signalised or controlled intersection. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade is 96 m. The survey recorded that 'The intersection does not meet clause 6.2 on either limb.'

The key relationship in this analysis is between 41 m and 96 m. The sight line was less than half the distance the truck needed to stop at the posted speed. At 37 km/h, the speed HT-407 was recorded at when the operator first sighted the light vehicle, 41 m gave about four seconds. That was not enough to stop, however alert the operator was and however quickly he responded. Attention, training and a radio call could each have affected whether a vehicle was in the intersection when a truck came over the crest. None of them could make the intersection stoppable once one was.

Nothing at the intersection compensated for the geometry. It had no warning sign, no give-way control, no delineation and no posted speed. Clause 6.4, which requires these, applies only to intersections shown on the plan's road layout drawing, and the RL 312 spur was not on it. The spur was opened under a works order with no management of change, which is discussed under Organisational Influences.

The hazard had already been reported in writing. On 04/08/2026 the dewatering technician wrote that 'you cannot see up the ramp from that road', and that 'I do not think they know the road is there.' He had met a descending truck he estimated at 'maybe sixty metres up'.

The sight line very likely contributed to both vehicles arriving at the intersection with neither operator aware of the other. The remaining uncertainty is that a short sight line only matters once a vehicle is in it. Whether a vehicle was in it depended on the other factors examined in this report.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

Dust was examined as a possible reason why the truck operator saw the light vehicle only at close range. The matter could not be resolved.

Some evidence pointed towards dust:

- The water cart had last passed that section 1 hour 37 minutes before the collision. The standing night shift interval is 90 minutes, so a pass was 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, recorded PM10 at 142 micrograms per cubic metre in the ten-minute interval covering the collision. The shift mean was 88.
- The truck operator described dust hanging on the ramp.
- The truck was descending behind its own lights into an unlit pit, which is the condition in which airborne dust most affects a driver.

Other evidence pointed away from it:

- The light vehicle driver had been at the same place about four seconds earlier. He was firm that the air was clear and that he could see the ramp surface and the windrow.
- The truck operator did not attribute any effect to the dust.
- The monitor reading was well below the site trigger of 250.
- The reading measured particulate at a monitor head 180 m away, not what could be seen from either cab. Nothing at Bindarra measures visibility.

The strongest point against dust is the sighting distance itself. The operator first saw the light vehicle at 41 m, which is the distance the survey showed the geometry allows in clear air. His account and the measured sight line therefore agree without dust having any effect.

The departure from the watering interval was real, and it occurred at the right place and time. Its effect on detection could not be established either way. Settling the question would require a measurement of what could be seen from the cabs on the night, and none exists. On the available evidence, it was about as likely as not that dust affected the truck operator's view. The matter remains unresolved, and nothing from it is carried into the findings.

The ramp surface was wet and the truck could not stop in the distance available

It was very unlikely that the ramp surface was wet. The met station recorded no precipitation, and the last rainfall was 26 days earlier. The investigator found the surface dry at 06:20 local. The truck also achieved a mean deceleration of 1.85 m/s² against its own tested dry-grade figure of 1.9 m/s². This possibility was therefore excluded.

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

The truck operator was on night 10 of a 14-night swing. The collision was at 02:47 local, 10 hours 47 minutes into the shift and within the period of lowest circadian alertness. Every limit in the contractor's fatigue management standard had been met. The operator's own sleep record showed 6 hours 30 minutes of sleep in the day before. It is virtually certain that this exposure existed.

It is unlikely that it contributed to this occurrence. Fatigue would have had to act through a slow or absent response. However, the recorded response of 1.4 seconds was within the 1.5 seconds assumed in the plan's stopping distance calculation and faster than the fleet figure. The braking

and steering that followed were immediate and in the only available direction. In addition, the geometry removed the margin a faster response would have needed.

The exposure is nonetheless built into the roster and worked on this and three other pits. At these pits, with no fixed lighting, the operator's own eyes are the only defence at an unlisted intersection. It is therefore likely that it increased risk.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

The pit channel was logged continuously, and the investigation reviewed the whole period from the shift's return from crib at 18:22:41 to about 2 minutes after the collision. Neither vehicle transmitted anything about the RL 312 spur, the dewatering bore, or a light vehicle being in the Stage 4 pit. Apart from logging on at 17:04, LV-118 made no transmission that shift. The truck operator, the light vehicle driver and the night shift supervisor were interviewed separately, and all three confirmed the absence of a call. It is virtually certain that no call was made.

No call was required. Clause 4.3.2 of BIN-TMP-004 states that at intersections not listed at Appendix B, vehicles give way under clause 4.1 and 'No call is required.' The RL 312 spur intersection was not listed. Each of the three people interviewed independently described the same practice: calls were made at the nine intersections on the crib room board and nowhere else. The supervisor stated that he had never corrected anyone for not calling at other intersections. He also considered that requiring calls at every intersection would make the channel unusable. The omission therefore did not depart from the plan or from local practice. It followed both exactly. The conditions that produced it, a positive call requirement tied to a list revised once a year, are analysed under Risk Controls.

It is likely that the absence of a call contributed to both vehicles arriving at the intersection with neither operator aware of the other. A call from either vehicle would have been heard by the other vehicle, by dispatch and by the supervisor. Dispatch was already tracking the truck closely enough to place it. At 18:46:18, it advised the excavator that 407 was 'about four minutes off you'. The assessment is not stronger because a call provides knowledge, not separation. Even with mutual awareness, the two vehicles would still have had to decide who went first across 41 m of sight line.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

The light vehicle driver's account was that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing, and drove out. When his account was read back to him, he corrected it from having stopped to having slowed, and said he did not believe he had come to a complete stop. That correction was against his own interest, and it agrees with the scene evidence: the impact point was 6.2 m into the ramp from the eastern edge of the spur mouth. It is virtually certain that the light vehicle entered the ramp without stopping and without establishing the truck's position.

This was not a departure from any control at the spur mouth, because there were none. The mouth had no give-way control, stop line, warning sign or posted speed, since clause 6.4 applies only to intersections on the road layout drawing. The driver described himself as applying the rule in clause 4.1, that light vehicles give way to heavy vehicles. He had not completed the Bindarra pit driving assessment and had never seen the traffic management plan. How he came to be driving in the pit without either is analysed under Organisational Influences.

Stopping would have changed little of what he could see. From the mouth, at his eye height of 1.45 m, the survey gave 38 m of sight line to a descending truck. The 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach. A look left from a stationary position would therefore have shown him approximately what his look at about 25 km/h showed him.

Without the light vehicle on the ramp there would have been no collision, and entering without establishing the truck's position was the step that placed it in the truck's path. It is very likely that this contributed to both vehicles arriving at the intersection with neither operator aware of the other. The assessment is not stronger because a full stop would have gained only a few metres of sight line, against a truck stopping distance of 96 m. The available sight distance at the intersection is analysed under Local Conditions.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

The fleet management system download recorded HT-407 at 37 km/h at 18:47:06, 41 m above the intersection. This was the last steady data point, and it corresponds to the distance at which the survey placed the operator's first line of sight to the spur mouth. From 18:47:07.4, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. The retarder was taken to maximum and 14 degrees of right steering was applied in the same movement. The response time of 1.4 seconds was within the 1.5 seconds assumed in the plan's own stopping distance calculation.

Mean deceleration to the collision was 1.85 m/s², against 1.9 m/s² recorded for the unit in its monthly brake test on 03/08/2026. The impact speed was 24 km/h; without braking it would have been 37 km/h. The scene evidence showed the braked scuff deviating 1.4 m to the right over its first 20 m, towards the wall side of the road. That was the only direction that did not carry the light vehicle towards a 100 m drop. The light vehicle came to rest against the inner windrow.

After stopping 19 m below the intersection, the operator reported the collision. He remained in the cab as dispatch directed until traffic was stopped, then went to the driver and stayed with him until the emergency response team arrived. It is virtually certain that these actions occurred, since they are shown in recorded data and corroborated by the scene marks.

It is very likely that the operator's response reduced the severity of the outcome. Braking lowered the impact speed on the side of the light vehicle, and the steering input placed the collision against the wall side of the ramp rather than the open edge.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

Every contributing chain in this investigation arrives at one condition: when the haul truck and the light vehicle reached the RL 312 spur intersection, neither operator knew the other was there. The analysis treats this as a single mechanism. The individual conditions that produced it are argued in their own subsections:

- the geometry that limited the sight line (Local Conditions)
- the suppressed collision awareness alert (Risk Controls)
- the absence of a positive call and the light vehicle's entry onto the ramp (Individual Actions)

The unresolved question of dust is also discussed under Local Conditions.

HT-407 was descending the main ramp empty at 37 km/h. Nothing on its approach indicated an intersection:

- The fleet management system in-cab map had last been published on 24/07/2026, a week before the spur was cut, and did not show the RL 312 spur.
- There was no warning sign on the approach.
- No call was made on the pit channel.
- The collision awareness system logged the light vehicle at 21.4 m but, inside the S4-LOAD geofence, raised nothing in the cab.

LV-118 was leaving the spur at about 25 km/h. It had 38 m of sight line up the ramp, no give-way control at the spur mouth and no call on the channel, and its driver had no knowledge of the truck. The night shift supervisor did not know a light vehicle was in the pit at all. No permit, call-in or other means existed by which he could have known.

The first sighting available to either operator was at 41 m, about four seconds before the vehicles met. That is what the survey predicts for the geometry of the intersection. The radio log, the truck's download, the collision awareness system log and the interviews with the truck operator, the light vehicle driver and the night shift supervisor were examined. None of them contains any indication that either operator knew of the other before that point.

One limitation qualifies this conclusion. LV-118 was not dispatched, carried no fleet management system and had only a passive collision awareness tag. Its movements before the final seconds of the approach therefore rest on the driver's own account alone. For that reason the investigation found it extremely likely, rather than virtually certain, that both vehicles arrived at the intersection with neither operator aware of the other.

The influence of this condition on the collision was also assessed as extremely likely. The collision followed directly and immediately from the convergence. The survey calculated the stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on this grade at 96 m. Against that figure, nothing that could happen after a first sighting at 41 m was capable of preventing the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The recorded data, the damage pattern and the debris field agree on how the vehicles met, and it is virtually certain that the collision occurred as follows:

- **Time and speed:** HT-407 struck LV-118 at 18:47:12.0 at 24 km/h, 4.6 seconds after the truck's brake application.
- **Position:** the impact was 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the toe of the inner windrow.
- **Force:** the truck's accelerometer logged 1.9 g longitudinal.
- **Damage:**
 - damage to LV-118 was concentrated on the right side from the B pillar forward
 - damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line
- **Debris:** right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The truck struck the side of the light vehicle, not its front. The spur joins the ramp from the east, on the inside of a left-hand bend, at an included angle of 71 degrees. A vehicle leaving the spur therefore crosses the path of descending traffic side-on, and this is the configuration the damage records. It placed the light vehicle's structure and its roll-over protection, rather than its front crumple zone, between the truck and the occupant.

The download shows that the truck operator's braking reduced the collision speed from the 37 km/h that would otherwise have applied to 24 km/h. That response is discussed under Individual Actions.

It is virtually certain that this impact contributed to the outcome, since it was the mechanism by which the convergence became an injury.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

The scene measurements, the vehicle examination and the medical information agree on what followed the impact:

- A continuous gouge ran 18.4 m down-ramp from the impact position.
- At 11.2 m the gouge changed from tyre scuff to body panel gouge, which marks where the vehicle went over.
- LV-118 came to rest on its left side against the inner windrow, facing down-ramp.

It is virtually certain that the light vehicle was pushed about 18 m and rolled onto its left side. It is virtually certain that this roll-over contributed to the injuries that make this a serious occurrence.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. These are the injuries a side impact and roll-over would be expected to produce when the vehicle structure holds and the restraint is worn, and two protective measures worked as intended:

- **Seatbelt:** the driver's seatbelt was found fastened, with loading marks on the webbing.
- **Roll-over protection structure:** it was deformed inward at the right rear mounting but was not breached.

With either of these absent, the outcome of the same sequence would very likely have been more severe.

Because the vehicle came to rest with the driver's door against the ground, the driver had to be extricated through the rear of the vehicle. The emergency response arrangements that governed the time this took are discussed under Risk Controls.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

This possibility was examined and found very unlikely. Brake performance tests on 03/08/2026 and 14/08/2026 gave 1.9 m/s² and 1.92 m/s², the occurrence download gave 1.85 m/s² between them, and no brake, retarder or park brake defect had been raised against the unit in the preceding twelve months.

Findings

From the evidence available, the following findings are made with respect to the collision between haul truck HT-407 and light vehicle LV-118 on the main ramp of the Stage 4 open pit at Bindarra Gold Mine at 18:47 UTC on 11/08/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

Contributing factors

- It is virtually certain that the collision awareness alerts were suppressed inside a geofence extended to cover the new intersection, and likely that this contributed. The contribution did not come from the advisory alert withheld at 18:47:09.2, which would have come after braking had begun. It came from fourteen months without an alert on that ramp, which had led the truck operator to take no alert as meaning a clear road. (Safety Issue)
- Both vehicles very likely arrived at the intersection with neither operator aware of the other, with the first sighting available to either at 41 m and about four seconds, and this very likely contributed directly to the collision.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, and this impact was, with virtual certainty, the mechanism by which the two vehicles' convergence became an injury.
- With virtual certainty, the light vehicle was pushed 18 m down the ramp and rolled onto its left side against the inner windrow, and this roll-over was, with virtual certainty, how the collision produced the driver's serious injuries.
- The traffic management guidance leaves the review trigger undefined and the interaction zones to the site; this discretion under MSG-14 existed with virtual certainty. On balance, it is likely that it contributed by leaving Bindarra's review, nomination and detectability arrangements to be filled in by the one site document, reviewed once a year. The guideline did not prevent the mine from reviewing its plan out of cycle or from listing every intersection, and Bindarra's plan met the guideline on its face. (Safety Issue)
- The RL 312 access spur was, with virtual certainty, cut and opened to traffic under a works order, with no management of change raised, and this very likely contributed. It removed the only route by which the new intersection would have been referred to the traffic management plan owner, the FMS road network and the collision awareness system configuration. (Safety Issue)
- The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m for a 220 t class truck empty at the posted speed; this is virtually certain, and it very likely contributed by leaving a truck at the posted speed unable to stop for anything at the intersection.
- Under BIN-TMP-004 it is virtually certain that positive radio calls were required only at listed intersections, and the list is rewritten once a year, with no provision to add an intersection between reviews or to refer a new road to the plan's owner. This very likely contributed by leaving the RL 312 intersection with no call requirement. (Safety Issue)
- Neither operator made a positive call on the pit channel before entering the intersection, which is virtually certain and was consistent with the plan and the local practice. It is likely that this contributed, although a call would have given each vehicle knowledge of the other rather than separation over 41 m of sight line.
- It is very likely that the contractor induction was accepted as equivalent, with the site supplement and pit driving assessment never verified, because the gatehouse issued access cards against the recognition agreement alone and the access card system had no

means of recording either item. It is likely that this contributed, although the give-way rule the assessment teaches was one the driver was already applying, and the intersection's sight line would have limited what attention alone could achieve. (Safety Issue)

- The light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, which is virtually certain, and this very likely contributed as the step that put the light vehicle across the descending truck's path. A full stop at the mouth would have given only a few metres more sight line and not the 96 m the truck needed.

Other factors that increased risk

- With virtual certainty, the night shift was rostered with one emergency response paramedic, with the second responder called from the village 11 km away, which delayed extrication by approximately 9 minutes. This did not contribute to the occurrence or its outcome, but it very likely increased risk because the site's night shift extrication capability does not scale with the severity of the casualty. (Safety Issue)
- The truck operator was, with virtual certainty, on night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low, within every limit of the contractor's fatigue management standard. The recorded 1.4 second response showed no sign that this contributed to the occurrence, but this roster pattern likely increased risk as a standing exposure worked by every haul truck operator in an unlit pit.

Other findings

- The ramp surface was very unlikely to have been wet. The met station, the surface as examined and the deceleration achieved each indicated a dry ramp, and a wet surface preventing the truck from stopping in the distance available was ruled out.
- The haul truck's service brakes were very unlikely to have been degraded, as brake tests before and after the occurrence and the occurrence download each showed performance to specification, and degraded stopping performance was ruled out.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting, which is virtually certain, and very likely reduced the severity of the outcome. It lowered the impact speed from 37 km/h to 24 km/h and kept the light vehicle against the inner windrow rather than the open edge.

Safety issues and actions

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

It is virtually certain that Mines Safety Guideline MSG-14 leaves three key terms for each site to define:

- **Review trigger.** It requires the traffic management plan to be reviewed after any significant change to the road network, but does not define a significant change.
- **Interaction zones.** It requires the operator to nominate the intersections and interaction zones where positive communication is required, but sets no minimum and no basis for the nomination.
- **Detectability.** It allows a light vehicle's detectability to be met by any one of five alternative means.

The plan is a site document. It is not approved by the Inspectorate, and is not submitted to it except on request after an occurrence.

Bindarra's plan met the guideline on its face, and the mine was not in breach. The issue is not a failure by the Inspectorate. It is that the discretion is wide enough for all three terms to be settled within one site document, reviewed once a year, with no outside review.

It is likely that this contributed to the occurrence. It left the site with no defined trigger or stated minimum that would have captured the eleven new Stage 4 roads never referred to the plan's owner. The investigation weighed this against contrary evidence. The guideline did not stop Bindarra from reviewing its plan out of cycle or from listing every intersection. On that view, the deciding weakness was internal, because nothing within the mine brought the question to the plan's owner. The investigation accepted the contribution on balance, with less confidence than applies to the site's own controls.

No safety action has been recorded in response to this safety issue.

Access spur cut and opened to traffic under a works order, with no management of change raised

The RL 312 access spur was cut and opened to traffic under works order WO-26-4471, with no management of change raised. CRL-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant is introduced. The works order cut 240 m of access to bore DW-14, classified the work as routine earthworks within the approved Stage 4 design and recorded management of change as not required under clause 5.3. Its handover field read N/A.

Clause 5.4 of the procedure sets out three referrals that a management of change record triggers:

- to the traffic management plan owner
- to the survey department for the fleet management system road network
- to the contractor's technical services group for the collision awareness system geofences.

Without the record, none of these referrals was made. The in-cab road network current on the night showed no intersection at RL 312. This issue very likely contributed to the occurrence, because the record was the only route by which the new intersection would have reached the plan, the in-cab map or the geofence configuration.

The classification on the front of a works order removed every consequence of building an intersection into a haul road. The issue extends beyond this spur. Fourteen new roads have been cut in Stage 4 since the last plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

No safety action has been recorded in response to this safety issue.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

It was extremely likely that the light vehicle driver's contractor induction was accepted as equivalent to the Bindarra induction, and that the site-specific supplement and the light vehicle pit driving assessment were never verified before he was issued an access card. CRL-STD-009 states that equivalence does not extend to either item. It also makes the engaging department responsible for verifying both before a card is issued.

The standard itself was sound. The shortfall was that the gatehouse, the only point where verification could take place, issued cards against the recognition agreement alone. It held no record of either item and was not asked for one, and nothing in the access card system distinguished a person who could drive in an operating pit from a person who could not. The driver's training file showed the Merrivale online module of 24/07/2026 and a 20 minute gatehouse briefing, with no supplement and no pit driving assessment. He had not seen BIN-TMP-004.

It was likely that this contributed to the occurrence. The pit driving assessment is where a light vehicle driver learns how the pit's traffic rules work in practice, including give-way, positive calls and what to do at a blind intersection. As a result, a person was driving a light vehicle on an active haul road on his third night at the mine without an assessment the site's own standard says may not be substituted, and no part of the access system was checking for it.

No safety action has been recorded in response to this safety issue.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Under the Bindarra Traffic Management Plan, BIN-TMP-004, a positive call on the pit channel was required only before entering one of the nine intersections listed at Appendix B. An intersection could reach that list only through the road layout drawing at Schedule 1, which, like the appendix, was reissued at the annual review. The plan had no provision for adding an intersection between reviews, and no requirement for a new road to be referred to the plan's owner when it was cut. The RL 312 spur appeared on neither document. The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine intersections on the crib room board and nowhere else.

The positive call was the mine's own defence against this kind of interaction, and at this intersection the plan's own terms left it inactive. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. The gap was not confined to one road. Eleven of the fourteen new Stage 4 roads cut since the last review had never been referred to the plan's owner. No out-of-cycle review had been held in the plan's eleven-year history. Four hazard reports describing the same interaction at four locations had each been closed with a reminder to the crew.

No safety action has been recorded in response to this safety issue.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet operated at Level 7, operator advisory, and did not raise alerts inside the S4-LOAD suppression geofence. That geofence was drawn around the loading face on 14/03/2025 to stop nuisance alerts. On 09/06/2026 the contractor's technical services group extended it northward as a system configuration task, to a boundary 180 m up-ramp of the RL 312 intersection. That change placed the intersection inside the suppressed area. The extension was not recorded in the traffic management plan and was not reviewed against it, and no person on site had been trained on the geofences.

The system detected the light vehicle at 18:47:09.2, at 21.4 m and closing at 13 km/h, and suppressed the alert. That alert would have come after the truck operator had begun braking. Its likely contribution lay elsewhere. The absence of alerts on that ramp over the preceding months had led the operator to treat silence as meaning an empty road, and he carried that expectation into the crest.

In the twelve months before the occurrence, 68 proximity events inside the geofence were suppressed. Nine of them involved a light vehicle tag on the main ramp below the boundary. None was reported, because the standing report covered alert counts and system health and did not report suppressions. The safety significance was not that a suppression geofence existed. It was that its boundary moved 180 m and nobody outside the group that moved it was told.

No safety action has been recorded in response to this safety issue.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. This is virtually certain. The second responder, who must be present before any extrication is begun, is called in from the accommodation village 11 km from the gatehouse.

On the night:

- dispatch paged the team at 02:48
- the first response vehicle entered the pit at 02:51 and reached the scene at 02:54
- the second responder arrived with the hydraulic rescue tool at 03:07
- the driver was extricated at 03:14, 27 minutes after the collision.

The site's post-incident review found that extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool.

The collision and the driver's injuries were complete before the emergency response was paged. The post-incident review found, on clinical grounds, that the delay did not affect the outcome. The injuries were stable and the driver was conscious throughout, restrained and protected by an intact roll-over structure. The arrangement did not contribute to the occurrence or its outcome.

It very likely increased risk nonetheless. The gap is structural rather than clinical. On every night shift, the site's extrication capability is a single person plus a drive from the village, and nothing in the arrangement changes with the severity of the casualty. A fuel fire, an unconscious casualty, or an airway that needed the vehicle opened within the first ten minutes could make the same delay of about nine minutes produce a different outcome.

No safety action has been recorded in response to this safety issue.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner

Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse